

Trade-offs for Performance in Reporting:

> What techniques can we use to make reports run faster, and what do we give up in return?

1. Indexes

Benefit

Reports become faster.

```
sql
CREATE INDEX idx_order_date
ON orders(order_date);
```

Trade-off

Faster SELECT
Slower INSERT/UPDATE/DELETE
More disk space

text

More Indexes

↓

Faster Reports

↓

Slower Writes

2. Materialized Views

Benefit

Store precomputed report data.

```
sql
CREATE MATERIALIZED VIEW sales_summary AS
SELECT region,
        SUM(amount)
FROM sales
GROUP BY region;
```

Trade-off

Very fast reporting
Data may become stale

text

Fast Report

↓

Old Data Until REFRESH

sql

```
REFRESH MATERIALIZED VIEW sales_summary;
```

3. Denormalization

Instead of joining many tables:

Normalized

text

Orders

Customers

Products

Report requires multiple joins.

Denormalized

text

Orders

- └ customer_name
- └ product_name

Trade-off

Faster reports

Data duplication

More storage

Update anomalies

4. Replicas (Read Replicas)

Benefit

Run reports on replica.

text

Primary DB

|

v

Replica DB

Trade-off

Reporting doesn't affect production

Replica lag may occur

text

Fast Reporting

vs

Slightly Old Data

5. Partitioning

Split large tables.

text
sales_2024
sales_2025
sales_2026

Trade-off

Faster queries on specific partitions
More complex administration

6. Aggregation Tables

Store summarized data.

sql
daily_sales

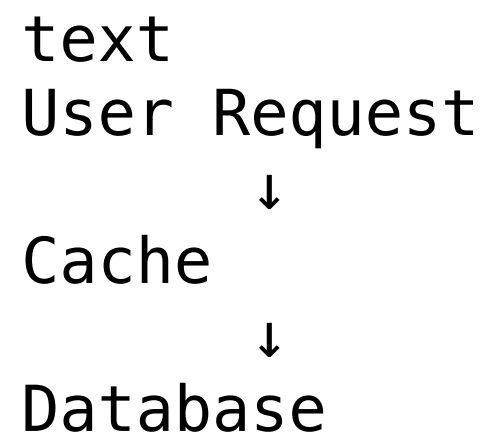
instead of calculating from millions of rows every time.

Trade-off

Fast reports
Extra ETL/maintenance

7. Caching

Store report results.



Trade-off

Very fast
Cache can become outdated

Technique	Gain	Trade-off
Indexes	Faster queries	Slower writes
Materialized Views	Faster reports	Stale data
Denormalization	Fewer joins	Data redundancy
Replication	Offload reporting	Replica lag
Partitioning	Faster scans	More complexity
Aggregation Tables	Quick summaries	Maintenance overhead
Caching	Instant response	Data freshness issues